

# **Runway Manager**

## User Manual

Version **1** July 20, 2026

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# 0 Administrative

## 0.1 Scope

This document serves as a guide on how Runway Manager operates, and how it can (not should) be used. The following topics are covered, non-exhaustively:

- Core functionality and features
- Details on the implementation of the A-CDM role
- User interface reference

This manual **does not** cover practical procedures such as ones covered by a facility's Standard Operating Procedures (SOP).

## 0.2 Definitions

- **RM** Runway Manager
- **Instance, Deployment** Specific deployed copy of, unless otherwise specified, Runway Manager. Each Runway Manager instance is hosted remotely and can be accessed through one or more clients.
- **Instance Manager** Person or group of persons responsible for managing a Runway Manager deployment.

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# 1 Introduction

Runway Manager is a sequence management tool that plays the A-CDM role in the CDM ecosystem.

## 1.1 System Architecture

Each RM deployment is assigned one "master" airport and is made up of the following components:

- A "Server", which manages and orchestrates the entire system, executing procedural logic and internal/external communication
- The "Clients" run on the user's machine, connect to the server and provide an interface for the user.

**Hosting** A Client may become the "host" and relay flight information to the server. There may only be one active host per instance. If a host connects when there is already another host connected, the previous host is overridden.

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## 2 The CDM Ecosystem

The CDM (Collaborative Decision - Making) model is an initiative launched by EuroControl aimed at improving cross-facility coordination. Its implementation leads to increased predictability and, thus, better planning across operations.

This section covers general information about the CDM model. See [A - CDM Implementation](#) for details on how Runway Manager implements the role of A-CDM in the CDM ecosystem.

### 2.1 Operational Milestones

Operational milestones are generic points in time applied to each flight in order to track and predict its progress. The CDM specification defines many such times. However, only the following are relevant in the scope of this project:

| Short | Long                         | Description                                                                                                                                               |
|-------|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| EOBT  | Estimated Off-Blocks Time    | Time filed in the flightplan for when the aircraft operator plans to be ready for startup. Does not reflect short-term delays.                            |
| TOBT  | Target Off-Blocks Time       | Time reported directly by the aircraft operator for when the flight will be ready for startup. Closely reflects short-term delays that may come up.       |
| TSAT  | Target Startup Approval Time | Time planned by ATC for the flight to receive startup clearance                                                                                           |
| ASAT  | Actual Startup Approval Time | Time at which the flight received start-up clearance                                                                                                      |
| TTOT  | Target Take-Off Time         | Time at which ATC plans for a flight to depart                                                                                                            |
| ATOT  | Actual Take-Off Time         | Actual departure ("wheels-up") time                                                                                                                       |
| CTOT  | Calculated Take-Off Time     | Commonly referred to as a "slot", a flight's CTOT can be thought of as a "target" TTOT, imposed by entities such as airport and A-CDM network management. |

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## 2.2 Components

The CDM system is made up of many parts (components). These can be grouped into two realms: the airport-local systems in the A-CDM (Airport-CDM) scope and the multiple Network Management Operations Center (NMOC) services, such as the Enhanced Tactical Flow Management System (ETFMS).

### 2.2.1 A-CDM

Conceptually, an Airport-CDM (A-CDM) system is local to its airport and separate from the network-wide NMOC infrastructure that lies above it.

An airport's A-CDM system is responsible for enforcing local procedures and creating the final departure sequence, while still taking into account information exchanged with NMOC facilities.

**Note** The term **A-CDM** refers only to the **airport-local** part of the CDM model. Sometimes, the term CDM is commonly used to refer to A-CDM, even though these are two (related, but) different concepts.

### 2.2.2 ETFMS

The Enhanced Tactical Flow Management System (ETFMS) facilitates improvements in flight management from the pre-planning stage to the arrival of the flight. It maximises the updating of flight-related data and thus improves the real picture of a given flight.

**DPI Messages** Departure Planning Information messages are sent from the A-CDM to the ETFMS in the order to supply the NMOC with flight data-related updates made available by sequencing tools and other airport management systems.

## 2.3 Flight Suspension

A suspended flight is barred from either start-up or departure, whichever comes earlier, until the suspension is lifted. A flight suspension is always attached to one of the following reasons:

- **NRA** (Not Reported Airport) — the flight has not departed within the allowable time frame.
- **MR** (Mandatory Route) — the flight is not complying with mandatory route constraints.
- **GS** (Ground Stop) — flight suspended due to a ground stop in force at its departure airport.

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## 2.4 ATFM-Exempt Flights

Some flights are exempt from Air Traffic Flow Management (ATFM) procedures and constraints. This means most CDM-related "features", like some operation milestones such as TSATs and TOBTs do not apply.

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## 3 A-CDM Implementation

This section covers Runway Manager's implementation of the A-CDM role. For general information on the CDM specification, see [section 2](#).

### 3.1 NMOC ("Slot Service") Integration

Runway Manager's NMOC integration system supports two different modes of operation:

- **Receive Only** — data is received, but nothing is sent.
- **Send and Receive** — data is received and sent (DPI messages).

#### 3.1.1 Supported DPI Messages

The following updates are sent by Runway Manager through [DPI Messages](#):

- TOBT updates
- Changes to earliest possible TTOT<sup>1</sup>
- A-CDM suspension state updates
- ARDT state changes

### 3.2 Flight Suspension / TTOT Cancellation

The cancellation of a flight's TTOT triggers an A-CDM suspension, and vice-versa.

A flight is also automatically suspended if:

- TSAT becomes later than 7 minutes past current time
- during initial processing, TOBT is later than current time
- an FLS-NRA suspension is received

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<sup>1</sup>earliest possible TTOT — TOBT/ASAT + Taxi Time

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## 4 Auto - Separation

Auto - separation **constantly checks and applies the best possible flight distribution** it can find, within the established limits, taking into account relative priority between flights.

Use of auto - separation is especially recommended during high - workload situations, as it requires almost little to no manual intervention.

**Note** Keep in mind that while it isn't unreliable, the auto - separation algorithm **may not always find the absolute best sequence** possible, which may cause very small inefficiencies. This is because the lead developer of the project (me) lacked the patience to look into actually formal and math - intensive papers regarding scheduling algorithms, so he made one up himself.

### 4.1 Minimum Separation Setting

Sets the minimum separation allowable between flights. Turning this all the way down to 0 **disables automatic sequencing** entirely.

### 4.2 Locked Flights

Locking a flight prevents it from getting automatically moved, except for:

- [queue delay](#)
- keeping locked flights from losing separation amongst themselves below the minimum separation setting, or below the initial separation at which they were placed, whichever is smaller.

### 4.3 Queue Delay

Flights with start - up clearance are kept from dipping below current time. This makes it so that accumulated delays "back the queue up" over time.

### 4.4 Priority Comparison

The following criteria are used to compare two flights and determine their relative priority and are sorted by order of importance. The comparison is terminated as soon as a criteria is met.

Flight, "A", has priority over another flight, "B", if:

1. A is locked and B is not;
2. A has an assigned CTOT and B does not;



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3. Both A and B are in the "[start-up sequence](#)" and A comes earlier;
  4. A is within its startup aproval window and is marked ready, and B is not;
  5. A's EOBT is earlier than B's;
  6. A's taxi time shorter than B's;
  7. A's logon time is earlier than B's;

**Start-up Sequence** The start-up sequence is a sorted list, calculated before each pass, that represents the order in which flights within their start-up aproval window or that have already received start-up aproval are sequenced. This list is used, as stated above, to preserve the sequence between such flights.

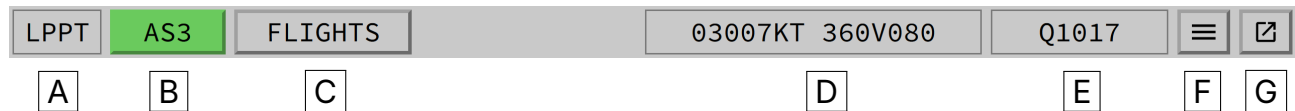
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## 5 Interface

### 5.1 Browser Client

The browser client can be accessed using a URL provided by the deployment manager.

#### 5.1.1 Top Bar



- A Airport ICAO identifier
- B Automatic Sequencing Setting (AS = Auto-Separation; SF = Auto-Shuffle)
- C Flight List Toggle. All flights, suspended or hidden, are listed here.
- D Current winds
- E Airport QNH
- F Options Menu
- G Pop-out Button (opens in a separate window)

The top bar may turn a **specific color to alert of a possible issue**:

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|               |                                                                                                                                                |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Red</b>    | No Euroscope host is set. The system <b>is not</b> receiving new flight information.                                                           |
| <b>Yellow</b> | An Euroscope host is set, however, its connection to the server may have timed out. The system <b>is not</b> receiving new flight information. |

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### 5.1.2 Flight Tag

|                    |          |      |     | OFF BLOCKS | ETOT/CTOT |       |  | SID     | QNH  |
|--------------------|----------|------|-----|------------|-----------|-------|--|---------|------|
| TAP1693 / A320 / M |          |      |     | 1602       | E1607     | 302   |  | OLBOD1X | 1017 |
|                    |          |      |     | 5          | LPMA      |       |  | 02      | 2066 |
| *                  | CALLSIGN | TYPE | WTC | TAXI TIME  | DEST.     | STAND |  | RWY     | SQK  |

**ETOT/CTOT** estimated departure. May display **C••••** if a CTOT is available.

**QNH** a flight's QNH is set upon receiving clearance, turning the field green. If it stops matching the airport's QNH, the field turns yellow.

**OFF BLOCKS**

1. If the flight has received start-up clearance, this field will show an ASAT in purple.
2. Otherwise, if a TSAT is available, its shown together with the resulting delay (TSAT - TOBT). Color depends on the delta between current time and TSAT:



3. If a TSAT isn't available the TOBT is shown in white.

The following actions can be triggered by interacting with a flight's tag:

| Field      | Button     | Action                                  |
|------------|------------|-----------------------------------------|
| *          | LMB (Drag) | Change <b>Departure Time</b>            |
| OFF BLOCKS | LMB        | Open <b>Off - Blocks</b> Menu           |
| STAND      | LMB        | Toggle <b>Flight Ready</b> Mark         |
| STAND      | RMB        | Open <b>Stand Selection</b> Menu        |
| QNH        | LMB        | Toggle <b>Clearance Tick</b>            |
| CALLSIGN   | LMB        | <b>Keep tag expanded</b> for 40 seconds |

### 5.1.3 Dialog Windows

#### Off-Blocks Menu (no ASAT)

TAP1693

PUSH

START-UP

TOBT

▲

1853

▼

SET NOW

SET NOW+30

OK

CLOSE

Set Push Status

Set Start-Up Status

TOBT

Set TOBT to current time

Set TOBT to current time + 30 minutes

#### Off-Blocks Menu (ASAT set)

TAP1693

ASAT

▲

1850

▼

OK

CLOSE

ASAT

#### Stand Selection Menu

TAP1693

STAND

▲

302

▼

OK

CLOSE

Stand Identifier

**Options Menu**

18:32:15      5ms

Euroscope...

[ON] Slot Service...

- Time / Latency
- Open Euroscope Authentication Menu
- Open Slot Service Mode window

**Euroscope Authentication Menu**

EUROSCOPE

HOST

● Carlos Mirpuri

TOKEN

.....

SHOW      COPY

GENERATE NEW TOKEN

CLOSE

- Hosting Controller Name
- Show token | Copy token
- Generate a new token
- Open Slot Service Mode Selection window

**Slot Service Mode Selection Menu**

SLOTSERVICE

MODE

▲

ON

▼

OK

CLOSE

Slot Service Mode

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## 5.2 Plugin

Bridges Euroscope to the Runway Manager stack, performing tasks such as uploading network flight data to the server (host) and displaying Runway Manager flight information in Euroscope.

Below is a list of commands supported by the plugin.

| Command          | Description                                                                                        |
|------------------|----------------------------------------------------------------------------------------------------|
| .rm              | Displays the Runway Manager window                                                                 |
| .rm host         | Start hosting                                                                                      |
| .rm up           | (Host Only) Pushes all flight data to Runway Manager at once                                       |
| .rm slave        | Stop hosting                                                                                       |
| .rm config       | Reload plugin config                                                                               |
| .rm p(ull)       | Pulls and applies flight data from Runway Manager to the network (ground state and clearance tick) |
| .rm hide         | Hide the Runway Manager window                                                                     |
| .rm auth <token> | Set authentication token                                                                           |

### 5.2.1 Loading

In Euroscope's **OTHER SET > Plug-ins** menu, load the plugin and make sure it can draw to all available radar screens and is first on the plugin list.

### 5.2.2 Authentication

Resembling Hoppie, Runway Manager uses a token-based system for authentication. To acquire your token, go to  > **Euroscope** and hit **COPY**. In Euroscope, use the plugin GUI or the ".rm auth" command to log in using this token. Upon login, the token is automatically saved for future sessions.

### 5.2.3 Connecting

When a controller connection is initiated, ATC Data is automatically pulled (tags are synchronized) if an active host is present.

### 5.2.4 Hosting

ATC data is automatically pulled if an active host was previously connected when you started hosting.